

The role of QA/QC in geotechnical instrumentation installation in TSF: critical control stages

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ABSTRACT

Geotechnical instrumentation provides the quantitative basis for stability assessments, risk management, and decision-making in tailings storage facilities (TSF). However, the reliability of monitoring data depends fundamentally on the conditions established during instrument installation, a phase where many installation aspects become irreversible once the borehole is sealed, backfill is placed, or facility construction advances. This paper identifies the key parameters that must be controlled and documented during the installation of geotechnical instruments in TSF, based on the authors' experience across multiple projects in Chile and Peru.

While the principles discussed apply to geotechnical instrumentation in general, the specific examples and parameters are illustrated using representative instruments commonly installed in TSFs through boreholes or excavation, including: vibrating wire piezometers (VWP), inclinometers, Shape Accel Array (SAA) systems, clino-extensometers (combined tilt and deformation sensors), and triaxial accelerometers. This work does not address post-installation data validation or historical record analysis.

Three critical control stages are defined: (1) pre-installation; (2) installation and completion; and (3) post-installation verification. Key parameters within each stage are classified by control stage and assessed for post-installation verifiability across representative instrument types. Installation quality deficiencies are observed systematically and are generally detected when monitoring data cannot be reconciled with expected physical conditions. Practical Quality Assurance and Quality Control (QA/QC) controls are proposed, centered on three principles: timely control of key parameters while they are still observable and identifiable; traceable documentation of each control point; and minimum acceptance criteria that must be satisfied before the installation stage is considered complete.

INTRODUCTION

The management of tailings storage facilities (TSF) throughout their life cycle requires reliable geotechnical monitoring data. Instruments such as vibrating wire piezometers (VWP), inclinometers, Shape Accel Array (SAA) systems, clino-extensometers, and accelerometers, among others, provide the quantitative basis for evaluating the critical variables of the main credible failure modes of a TSF, including pore water pressures, deformations, and seismic response. The Global Industry Standard on Tailings Management (GISTM) establishes in Principle 7 the requirement to design, implement, and operate comprehensive monitoring systems for risk management (ICMM, UNEP & PRI, 2020).

Existing literature extensively addresses instrument selection, monitoring system design (Dunnicliff, 1993), installation methods, and data interpretation, but comparatively little addresses the Quality Assurance and Quality Control (QA/QC) of the installation process itself as a system of controls exercised before, during, and after execution.

The GISTM framework introduces Construction versus Design Intent Verification (CDIV), which for geotechnical instrumentation should verify that each instrument is installed per engineering specifications, manufacturer requirements, and design intent. Good practice guidance further emphasizes that construction records, including QA/QC documentation, topographic surveys, and as-built drawings, must demonstrate compliance with technical specifications (ICMM, 2025). In practice, this is often incomplete or delegated to personnel without sufficient technical expertise. This paper addresses the installation process, focusing on parameters that become irreversible or unverifiable once that phase is complete.

METHODOLOGY

The findings presented are based on the authors' direct experience in geotechnical instrumentation across at least 14 TSF projects in Chile and Peru since 2015, covering construction, operation, and closure phases, and involving more than 400 instruments including VWP, inclinometers, SAA systems, clino-extensometers, and accelerometers, among others. This experience included direct observation during instrumentation campaigns, review of installation record packages and calibration certificates, commissioning and reception activities, and technical assessment of monitoring systems for risk management purposes. The key parameters and QA/QC controls presented here were identified through recurrent observations of installation deficiencies encountered during installation supervision, documentation review, commissioning, audits, and data interpretation. Each parameter was classified according to its control stage and its post-installation verifiability, defined as the degree to which the parameter can be independently determined in the absence of installation records once the instrument is in place (see Table 1).

RESULTS AND DISCUSSION

Limitations of Post-Installation QA/QC

Many key parameters associated with geotechnical instrumentation, such as the initial reading, installation elevation and depth, sensor orientation, annular fill characteristics, and compliance with manufacturer-specified procedures, are physically defined during installation. Once the borehole is sealed, grouted, or backfilled, these variables can no longer be independently verified with certainty.

Current QA/QC frameworks typically assign QC responsibilities to construction and instrumentation contractors through self-inspection, while QA is performed separately, reviewing documentation and validating record packages after installation. The fundamental problem is that QA review often engages minimally or late, when physical conditions can no longer be observed, leaving the reviewer dependent on records generated by the installing parties, with limited ability to validate them.

Installation deficiencies are typically not detected during record package review or commissioning; they generally become apparent when monitoring data cannot be reconciled with expected behavior. At that point, installation conditions can no longer be reconstructed, and affected instruments must be used with restrictions, limited confidence, or discarded, representing a loss of investment and monitoring coverage critical for TSF risk management.

Two recurrent examples illustrate this pattern. In the first, VWP showed stable positive pressure readings over extended periods, while a Casagrande piezometer in the same borehole remained dry; without traceable elevation records and documented initial readings, the discrepancy could not be resolved, and the instruments were downgraded to limited-confidence status. In the second, during commissioning, the initial readings configured in some of the VWP dataloggers were found to be incorrect, a deficiency detected only because the reviewer questioned the physical coherence of the readings; without this check, the error would have gone undetected.

Critical QA/QC Control Stages During Installation

Pre-installation

This stage covers activities from instrument reception through placement in the final design position. Critical controls include: verifying that calibration certificates and technical specifications match design requirements; confirming that all accessories and spare parts have been supplied; executing functional tests to verify that each instrument is operational within expected parameters; and, for VWP, documenting filter saturation procedures per manufacturer specifications, including duration and method (immersion, vacuum, or other).

Calibration certificates are generally available with the supply; however, records of pre-installation procedures, functional tests, and reception verification are frequently incomplete or nonexistent, with significant consequences since these processes cannot be repeated once installed.

Installation and completion

This stage covers the physical emplacement of the instrument, including borehole completion (annular fill, grouting, casing installation, or push-in method), trench excavation and backfill, and construction of support infrastructure (datalogger posts, solar panels, perimeter protection). Most irreversible controls occur at this stage, including: timely topographic survey of the borehole collar (elevation and coordinates) immediately after installation, before losing this reference due to construction activity; accurate measurement of instrument depth and monitored stratum; control of the annular fill (materials, placement method, and layer thicknesses); recording of initial readings with methodology, environmental conditions, and instrument status; for fully grouted VWP (Contreras, Grosser & Ver Strate, 2008), verification that grout column pressure does not exceed the instrument's allowable over-range (Mikkelsen & Green, 2003); for VWP with sand and gravel filter, validation that sensor elevation is consistent with the backfill; for inclinometer casings, measurement of groove orientation and torsion; for accelerometers, documentation of axis alignment relative to the dam crest; and preparation of as-built sketches.

Several of these controls are routinely omitted or inadequately documented: topographic surveys are frequently delayed, sometimes losing the borehole collar reference; grout column height is rarely verified against the instrument's pressure rating; torsion and axis orientation records are often absent or ambiguous; and initial reading methodology is seldom clearly recorded.

Post-installation verification

This stage covers the period immediately following installation, before the instrument is formally accepted, and is the last opportunity to detect problems before correction or reinstallation becomes impractical. Critical controls include: verifying that calibration factors and conversion parameters loaded into the datalogger or platform match the instrument's calibration certificate; a basic physical coherence check of whether the readings make sense given installation depth and the expected behavior of the monitored variable (groundwater level, deformations, and/or accelerations); and verification of connectivity and data transmission.

Datalogger configuration errors are not infrequent: calibration factors from one instrument applied to another, incorrect unit conversions, wrong equations, or reference offsets that do not match the documented initial reading. These are deficiencies detectable through timely QA/QC review and physical coherence checks, as illustrated in the second example above.

Key Installation Parameters and Post-Installation Verifiability

Table 1 presents the key parameters identified, organized by control stage and assessed for post-installation verifiability by representative instrument type, reflecting the degree to which each parameter can be independently determined in the absence of installation records. This classification serves as a practical reference for prioritizing QA/QC controls during installation.

Table 1 Key installation parameters: control stage and post-installation verifiability

Key parameter	VWP	Inclinometers / Clino-extensometers	Accelerometers
Pre-installation			
Verification of calibration certificates and accessories	Possible	Possible	Possible
Pre-installation functional test	Impossible	Impossible	Impossible
Filter saturation	Impossible	N/A	N/A
Installation and completion			
Topographic survey (elevation, coordinates)	Limited	Limited	Possible
Sequence and control of annular fill	Impossible	Impossible	N/A
Over-range pressure check ¹	Impossible	N/A	N/A
Sensor axis orientation	N/A	Limited	Possible
Casing torsion (surface measurement)	N/A	Limited	N/A
Initial readings	Impossible	Impossible	N/A
Post-installation verification			
Datalogger/platform configuration check	Possible	Possible	Possible
Physical coherence check	Possible	Possible	Possible

Notes: ¹Applies to fully grouted VWP. Definitions: N/A = parameter does not apply to that instrument type. Post-installation verifiability assessed in the absence of installation records: Impossible = cannot be determined retrospectively; Limited = can be estimated indirectly with uncertainty; Possible = verifiable after installation.

PROPOSED QA/QC APPROACH

Timely Control and Documentation

Key parameters must be controlled while still observable and measurable, requiring specialist QA/QC personnel present during installation. This specialist must be independent of the construction and instrumentation contractors, with demonstrated experience in the instrument types installed and the ability to evaluate whether control readings are physically coherent. Each critical control must be documented with the date, the person performing and witnessing it, the verification result, and supporting evidence, generated contemporaneously with execution rather than reconstructed later.

Minimum Acceptance Criteria

The installation of an instrument should not be considered accepted until: the initial reading was acquired under documented and verifiable conditions; elevation, depth, and coordinates were surveyed in a timely manner; completion conditions were executed and documented per specifications; and a check confirmed that the readings are physically coherent. If any condition

cannot be satisfied, the non-conformity must be formally recorded, and a decision made on whether the instrument is discarded, used with restrictions, or maintained for reference only.

The minimum record package should include, per instrument: calibration certificate and functional test records; filter saturation record (VWP); surveyed collar elevation and coordinates; annular fill or completion log; initial reading record with methodology and conditions; casing torsion and groove orientation record (inclinometers and clino-extensometers); sensor axis alignment record relative to dam crest axes (accelerometers); and datalogger configuration verified against calibration certificate.

The proposed framework supports GISTM implementation. Timely, independent control enables CDIV to identify discrepancies between as-built conditions and design intent while still observable, consistent with critical control management practice, which emphasizes physical verification by parties independent of those executing the work (ICMM, 2026). Traceable documentation and minimum acceptance criteria further help ensure that the monitoring system relies on instruments whose installation quality can be demonstrated, supporting Principle 7 (ICMM, UNEP & PRI, 2020).

CONCLUSIONS

QA/QC for geotechnical instrumentation installation must focus on controlling and documenting key parameters during the installation process itself, not on post-installation documentation review, since many of these parameters become unverifiable once the instrument is buried, grouted, or sealed.

Three critical control stages have been identified: pre-installation, installation and completion, and post-installation verification. Each contains parameters that, if not properly controlled, compromise confidence in monitoring data. These deficiencies have been observed systematically and are typically detected when data fails to provide meaningful physical interpretation.

The proposed approach, based on timely control by specialized QA/QC personnel, contemporaneous and traceable documentation, and minimum acceptance criteria, offers a practical framework to close the gap between current practice and GISTM requirements, and strengthens the monitoring system upon which the performance assessment of the TSF depends.

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